Experiment 03 – Lab 3

Precision Prompting for Information Extraction

Aim

To use ChatGPT to extract structured information from text and present it in JSON or tabular format, then evaluate the accuracy and format of the output.

Theory

Precision prompting means giving clear instructions about what information to extract and how to format it. Specifying fields, output format, and rules for missing information helps make the response easier to check and reuse.

Task 1: Extract Contact Information

Source text: Priya Sharma is a student. Her email is priya.sharma@example.com and her phone number is 9876543210. She is studying Artificial Intelligence and Data Science at MMIT Pune.

Prompt used: Extract the person's name, email, phone number, course, and college from the text. Return only valid JSON with keys name, email, phone, course, and college. Do not add information that is not present in the source.

ChatGPT output (JSON):

```
{
  "name": "Priya Sharma",
  "email": "priya.sharma@example.com",
  "phone": "9876543210",
  "course": "Artificial Intelligence and Data Science",
  "college": "MMIT Pune"
}
```

Accuracy and format evaluation: All five fields match the source text. The output uses valid JSON-style key-value pairs and includes no extra fields. The phone number is stored as a string to preserve its digits.

Task 2: Extract Keywords from a Resume Summary

Source text: A second-year student with knowledge of C++, Python, data structures, algorithms, and basic SQL. Completed a mini-project on a student record management system and is interested in software development.

Prompt used: Extract the programming languages, technical skills, project, and career interest from the resume summary. Return a table with columns Category and Extracted Information. Use only details explicitly stated in the text.

Category	Extracted Information
Programming languages	C++, Python
Technical skills	Data structures, algorithms, basic SQL
Project	Student record management system
Career interest	Software development

Accuracy and format evaluation: The extracted categories are supported by the source text. The table has the two requested columns and groups related details clearly. “Basic SQL” is retained as stated rather than expanded into an unsupported proficiency claim.

Task 3: Handling Missing Information

Source text: Rahul Patil is learning Python and C++. His email address has not been provided.

Prompt used: Extract name, programming languages, and email into JSON. If a field is missing, use null. Do not guess missing values.

ChatGPT output (JSON):

```
{
  "name": "Rahul Patil",
  "programming_languages": ["Python", "C++"],
  "email": null
}
```

Accuracy and format evaluation: The name and languages match the source. The missing email is represented as null rather than invented, following the prompt's instruction. The response follows the requested JSON structure.

Output Accuracy Checklist

Check	What to verify	Result
Completeness	All requested fields are present	Pass
Correctness	Values match the source text	Pass for examples above
No guessing	Missing data is not invented	Pass
Format	JSON/table follows requested structure	Pass

Conclusion

Precision prompting can help ChatGPT extract information in structured formats such as JSON and tables. Checking extracted values against the original text helps assess accuracy, while clear formatting instructions make the output easier to use. Missing information should be marked clearly rather than guessed.

Result: Prompts were designed to extract contact information and resume details, and the outputs were evaluated for accuracy and format.